

Math 252

Assignment 1

The assignment is due on October 4th.

1. Solve the initial-value problem

$$\frac{dy}{dx} = \frac{xy^4}{3\sqrt{1+x^2}}, \quad y(0) = 1.$$

2. Solve the differential equation

$$x^2 \frac{dy}{dx} + 3xy = e^{2x}.$$

3. Solve the differential equation

$$(3x^2y^2 + e^{2y} + 4x^3) dx + (2x^3y + 3\cos(3y) + 2xe^{2y}) dy = 0.$$

4. Solve the differential equation

$$\frac{dy}{dx} = \frac{xy + y^2}{x^2}.$$

5. Solve the differential equation

$$x \frac{dy}{dx} + 2y = x^3y^2 \sin x.$$

6. Find an explicit solution (i.e., solve for y) of the following IVP.

$$\frac{dy}{dx} = (1 + x + y)^2, \quad y(0) = 0.$$

7. The rate of change in the intensity I of light below the surface of the ocean with respect to the depth y is proportional to I . Assume that the intensity of light at the surface is I_0 , i.e., $I = I_0$ when $y = 0$. If the intensity at $y = 5.0\text{m}$ is 50% of I_0 , at what depth y is the intensity 15% of I_0 ? Set-up and solve a differential equation to answer the question.
8. A tank contains 2000 L of pure fresh water initially. Brine containing 15 g of salt per liter is entering the tank at a rate of 40 L/min; the well-mixed solution is pumped out at the same rate. Assume that the initial fresh water in the tank contained no salt.
- Find the mass $m(t)$ of salt in the tank as a function of time t .
 - After how much time do we have 10 kg of salt in the tank?
 - How much salt will there be in the tank in the long run, i.e., as $t \rightarrow \infty$?